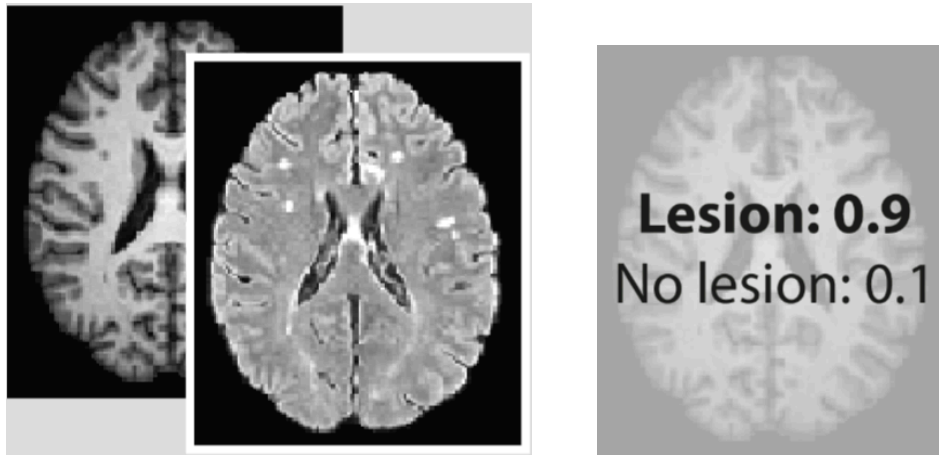


Use Case 3: Identification of new multiple sclerosis lesions in follow-up MRI scans



Problem category: Image Level Classification

Recommended metrics:

- Multi-class Counting Metric: [Expected Cost \(EC\)](#)
- Per-class Counting Metric: [Fbeta Score](#)
- Multi-threshold Metric: [Average Precision \(AP\)](#)
- Calibration Metrics: [Brier Score \(BS\)](#)

Fingerprint answers:

Fingerprint	Answer
Decision rule (cutoff method) applied to predicted class scores	Argmax-based
Penalization of errors: Unequal severity of class confusions	Yes
Costs for class confusions available	Yes
Provided class prevalences reflect the population of interest	Yes
Availability of predicted class scores	Yes
Calibration assessment requested	Yes
Comparative calibration assessment requested	Comparison of overall performance across classifiers
Interpretable estimate of calibration error requested	Joint assessment of calibration and discrimination

Decision between metrics:

- [Multi-class Counting Metric] Weighted Cohen's Kappa (WCK) vs. Expected Cost (EC)
 - Answer: Expected Cost (EC)
- [Per-class Counting Metric] Positive Likelihood Ratio (LR+) vs. Sensitivity vs. F_β Score
 - Answer: F_β Score
- [Mutli-threshold Metric] Area under the Receiver Operating Characteristic Curve (AUROC) vs. Average Precision (AP)
 - Average Precision (AP)
- [Calibration Metric]: Brier Score (BS) vs. Negative Log Likelihood (NLL)
 - Answer: Brier Score (BS)